Parameter-Efficient Quantum Sentiment Classification: A Controlled Empirical Study of DisCoCat on SST-2

Nhat-Linh Do[1], Quang-Thai Le[1], and Minh-Tan Tran-Huynh[2]

[1] Faculty of Business Data Science, Banking University of Ho Chi Minh City, Vietnam
`{linh.do,thailq}@hub.edu.vn`
[2] Faculty of Management Information Systems, Banking University of Ho Chi Minh City, Vietnam
`tanthm@hub.edu.vn`

Abstract. We investigate the application of Distributional Compositional Categorical (DisCoCat) models with variational quantum circuits to binary sentiment classification on short phrases (2–5 tokens) drawn from the Stanford Sentiment Treebank (SST-2). Each phrase is converted into a DisCoCat string diagram using the `lambeq` library, parameterised by a quantum ansatz, and trained on the `PennyLane` simulator. A controlled experimental grid of 60 quantum runs is conducted, spanning two diagram readers (Spiders, Cups), two ansatz families (IQP, Sim14), three circuit depths and five random seeds. Three classical baselines—Logistic Regression with TF–IDF bigrams, Logistic Regression with averaged GloVe-50d embeddings, and a single-layer bidirectional LSTM—are evaluated on the same balanced 5,000/400/400 split.

The Spiders reader (which attains identical accuracy across all six ansatz–depth configurations, as documented below) achieves 92.05%±0.64% test accuracy using only 2,940 trainable parameters. This result is statistically indistinguishable from the BiLSTM baseline (91.83% ± 0.42%; McNemar $p = 0.60$), which requires 28 times as many parameters (82,369), and trails the strongest classical baseline (LR + TF–IDF, 94.25%, 2,829 parameters; McNemar $p = 0.045$) by a small but statistically detectable margin of 2.2 percentage points. Two methodologically informative observations are reported. First, with single-qubit atomic types the Spiders reader becomes representationally degenerate: every ansatz–depth combination reduces to the same single-qubit unitary, and all six Spiders configurations attain identical accuracy. Second, the Cups reader combined with the IQP ansatz exhibits a reproducible barren-plateau collapse at depth two (62.8% across all five seeds), which persists despite the use of a `ReduceLROnPlateau` scheduler and gradient clipping; the Sim14 ansatz remains stable across the same depth sweep (91.85–92.10%). Code and experiment artefacts are available on request.

Keywords: Quantum Natural Language Processing · DisCoCat · Pregroup Grammar · Variational Quantum Circuits · SST-2 · Sentiment Classification · `lambeq` · PennyLane · Barren Plateau.

1 Introduction

Natural Language Processing has been transformed by large pretrained language models, but these models require enormous parameter budgets that are impractical on current quantum hardware. Quantum Natural Language Processing (QNLP), based on the categorical compositional distributional model DisCoCat [3], offers a structurally different paradigm in which sentence meaning is built by composing word tensors according to the pregroup grammar of the sentence. Recent practical results from Quantinuum [9,6] have demonstrated that small DisCoCat models can be trained on real near-term quantum hardware, suggesting that QNLP is a promising research direction even in the NISQ era.

1.1 Problem Statement

We focus on the canonical task of *binary sentiment classification* on short phrases (2–5 words) drawn from the Stanford Sentiment Treebank [16]. The dataset is restricted to a balanced split of 2,500/2,500 train, 200/200 dev and 200/200 test (5,000/400/400 phrases in total) over a 1,000-token vocabulary, ensuring that quantum simulation is tractable while retaining enough data for stable statistical comparison.

1.2 Contributions

The contributions of this paper are threefold. First, we release a fully reproducible end-to-end pipeline from raw Stanford SST data, through DisCoCat diagrams and parameterised quantum circuits, to final evaluation metrics. Second, we report a 60-run controlled experimental grid covering two readers, two ansatzes, three depths and five seeds, with means and standard deviations for each configuration, and compare the results against three classical baselines—LR with TF–IDF bigrams, LR with averaged GloVe embeddings, and a single-layer BiLSTM—evaluated on identical train, development and test splits.

Two methodologically informative findings emerge from this grid. With single-qubit atomic types, the Spiders reader collapses every ansatz and depth configuration to the same single-qubit unitary, rendering Spiders unsuitable for ansatz ablation studies unless higher-dimensional atomic types or a grammatical CCG reader is employed. The Cups reader combined with the IQP ansatz at depth two falls into a barren plateau that persists under a learning-rate scheduler and gradient clipping, providing a concrete empirical instance of the phenomenon described by McClean et al. [10].

In addition, we document an infrastructure-level issue: the BobcatParser model distributed by `lambeq` 0.5.0 references the deprecated `qnlp.cambridgequantum.com` domain (no longer operational following the merger of Cambridge Quantum into Quantinuum), and the model is therefore not retrievable through the standard library interface. A workaround based on the self-contained Spiders and Cups readers is described in Section 3 and discussed further in Section 6.

1.3 Paper Structure

Section 2 reviews DisCoCat, QNLP, and the variational quantum ansatzes used in this work. Section 3 describes the data pipeline, the quantum model, the classical baselines, and the evaluation protocol. Section 4 details the experimental setup. Section 5 presents the main results, including the per-length analysis and the ablation study. Section 6 interprets the findings and lists limitations. Section 7 concludes and points to future work.

2 Background and Related Work

2.1 DisCoCat: Compositional Distributional Semantics

DisCoCat [3] unifies distributional semantics (word meanings as vectors) with compositional grammar (sentence structure as morphisms in a monoidal category). Pregroup grammar [7] assigns each word a *pregroup type* formed from a small set of atomic types (e.g. n for noun, s for sentence) together with right/left adjoints. A sentence is grammatical if its type sequence reduces to s under the cup contractions; the resulting reduction is represented as a string diagram that simultaneously encodes the syntax and the semantic composition.

In the categorical formulation, each word is a morphism from the unit object to a tensor product of (possibly adjoint) atomic types, and the sentence is a composition of these morphisms. Realising this composition in a *finite-dimensional Hilbert space* immediately suggests an implementation on a quantum device: words become parameterised quantum states, atomic types become qubit registers, and grammatical contractions become measurement-induced post-selection or measurement of an output register.

2.2 Quantum NLP and the `lambeq` Toolkit

`lambeq` [6] is a high-level Python library that implements the full DisCoCat pipeline: parsing text into a CCG derivation, converting the derivation into a pregroup diagram, applying optional rewriter rules, and assigning a variational quantum circuit through a chosen *ansatz*. Multiple *readers* are provided. A *Bobcat* neural CCG parser produces full grammatical diagrams; simpler readers such as the *Spiders* and *Cups* readers produce structurally smaller diagrams that do not require parser inference and are commonly used as baselines. More recent applied QNLP work includes the quantum recurrent architectures of Xu et al. [17] for text classification at scale, and the quantum transfer-learning study of Guarasci et al. [4] on acceptability judgements; both support the present line of research with comparable QNLP results on public benchmarks.

2.3 Variational Quantum Ansatzes

A quantum ansatz is a parameterised circuit family with a fixed topology but trainable rotation angles. The IQP family [5] alternates single-qubit Hadamards with diagonal phase entanglers. The Sim14 family [15] is a hardware-efficient ansatz with controlled rotation gates. The StronglyEntangling family is widely used as a baseline in PennyLane [1]. The number of layers controls expressive capacity but also the trainability under barren plateaus, a phenomenon whose theoretical foundations were laid out in [10] and have been substantially refined in the recent review [8] and the Lie algebraic characterisation of [13]. Broader framings of when quantum machine learning offers genuine practical advantage appear in [2,14].

2.4 The SST-2 Dataset

The Stanford Sentiment Treebank (SST) [16] provides fine-grained sentiment annotations on phrases extracted from parse trees of movie reviews. We restrict to the binary subset (*SST-2*) by thresholding the continuous sentiment score: phrases with score below 0.4 are labelled negative, phrases above 0.6 are labelled positive, and neutral phrases in the band $[0.4, 0.6]$ are discarded. The full SST phrase pool contains roughly 50,000 phrases of length at most five.

3 Methodology

Figure 1 summarises the end-to-end pipeline of this work. An input phrase is tokenised, converted to a DisCoCat string diagram by a `lambeq` reader (Spiders or Cups), parameterised by an ansatz (IQP or Sim14) at a chosen depth, executed on the `PennyLane` simulator, and the resulting two-class probability vector is argmax-decoded to a `POS`/`NEG` label.

3.1 Dataset Preparation

We download the raw Stanford SST data and join the four files `datasetSentences.txt`, `datasetSplit.txt`, `dictionary.txt` and `sentiment_labels.txt` to obtain a DataFrame with the columns *text*, *score*, and (sentence-level) *split*. Because the phrase-level split in the original SST data is restricted to sentence root phrases, we replace it with a stratified random split over the full phrase pool.

The preprocessing pipeline (Table 1) is as follows. We first restrict to phrases with at least two and at most five tokens, then binarise the sentiment by the thresholds above. Following standard QNLP practice we lowercase all text. To control the size of the parameter space of the quantum model we keep only the top-K most frequent training tokens ($K = 1000$) and discard any phrase containing a token outside this vocabulary. Finally we apply *class-balanced sampling*: we draw 1,500 positive and 1,500 negative training phrases at random, together with 200/200 for development and 200/200 for test.

We adopt a two-stage sampling protocol to ensure that test vocabulary is a subset of training vocabulary: (i) sample the 5,000 balanced training phrases first, then (ii) restrict the dev/test pools to phrases whose words all appear in the sampled training set before drawing the final 400 dev and 400 test phrases.

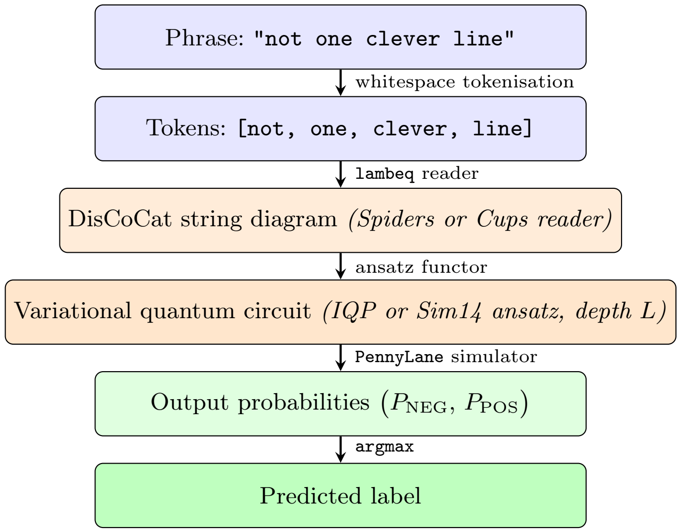

Fig. 1. End-to-end QNLP pipeline for binary sentiment classification on short phrases. The two methodological choices ablated in this work (highlighted in orange) are the *reader* that produces the DisCoCat string diagram, and the *ansatz* that parameterises the diagram into a quantum circuit. All other stages are fixed across experiments.

Without this constraint, the quantum model would have no learned parameters for words appearing only in the test split, since each DisCoCat box corresponds to a distinct trainable tensor.

3.2 DisCoCat Pipeline

We use `lambeq` version 0.5.0. We initially intended to use the BobcatParser, but the model URL hosted on `qnlp.cambridgequantum.com` is no longer reachable following the Cambridge Quantum/Quantinuum merger; this is a notable reproducibility issue discussed further in Section 6. As a workable substitute we use two non-parsing readers:

`SpidersReader` assigns the sentence type s to every word and merges them with a Frobenius spider (cf. Fig. 2). The resulting diagram has a single output wire s and is invariant under permutation of the input words.

`CupsReader` inserts a `START` token of type s and assigns type $s.r \otimes s$ to each subsequent word, then contracts the adjoints via cups (cf. Fig. 3). The resulting diagram retains the linear order of tokens and has higher composition depth than the Spiders reader.

For both readers we set the atomic type dimensions to one qubit each ($q_N = q_S = 1$), following the convention of Lorenz et al. [9]. The rewriter rules for prepositional phrases, determiners, and auxiliary verbs apply only to CCG-derived diagrams and are therefore disabled.

Table 1. Preprocessing pipeline producing the `qnlp` subset.

Stage	Train	Dev	Test
Raw phrases (length ≤ 5)	37,883	4,735	4,737
After $2 \leq \text{len} \leq 5$	33,399	4,195	4,162
After lowercase + vocab-1000 + OOV filter	9,711	1,172	1,216
After class-balanced sampling (train first)	**5,000**	—	—
After word-coverage filter (dev/test $\subseteq$ train words)	—	935	1,018
After class-balanced sampling (dev/test)	—	**400**	**400**
Positive / negative ratio (final)	2,500:2,500	200:200	200:200
Mean phrase length (final, tokens)	3.15	3.20	3.30

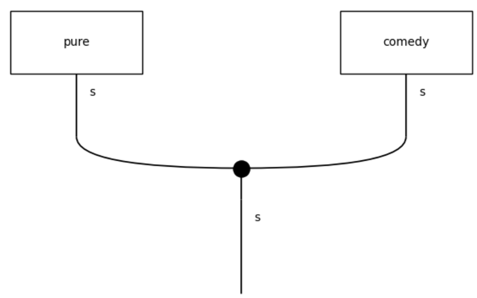

Fig. 2. Spider diagram for the phrase *pure comedy*: each word box has output type s and the two boxes are merged by a Frobenius spider into a single output wire of type s.

3.3 Quantum Models

The diagrams are converted into parameterised quantum circuits using either the IQP [5] or the Sim14 [15] ansatz with depths $L \in \{1, 2, 3\}$. The full circuit is then wrapped in a `PennyLaneModel` that exposes the trainable parameters as a PyTorch `nn.Module`. Simulation is performed on the `lightning.qubit` backend.

For binary classification the model outputs a length-2 probability vector over the output register $|0\rangle, |1\rangle$; we train it with the cross-entropy loss against the one-hot labels.

3.4 Classical Baselines

We evaluate three classical baselines using the *same* 5,000/400/400 split:

LR + TF–IDF A logistic regression with unigram and bigram TF–IDF features, $C \in \{10^{-2}, \ldots, 10^{2}\}$ tuned on the dev set, using `liblinear`.

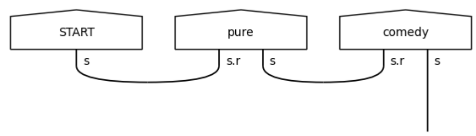

Fig. 3. Cups diagram for the phrase *pure comedy*: the `START` token initialises a sentence wire of type s and the right adjoints of successive word types are contracted via cups, leaving a single s wire at the bottom.

LR + GloVe A logistic regression on the mean-pooled GloVe-50d [12] word vector of the phrase, with the same hyperparameter grid.

BiLSTM 1L A randomly initialised embedding of dimension 32, a 1-layer bidirectional LSTM with hidden size 64, mean-pooling over valid positions, dropout 0.3, and a linear classifier. Trained with Adam (lr $= 10^{-3}$) and BCE loss, early stopping on dev loss with patience 5.

3.5 Training and Evaluation Protocol

All quantum models are trained with Adam (lr $= 10^{-2}$, weight decay 10^{-4}) and gradient clipping at unit norm. A `ReduceLROnPlateau` scheduler halves the learning rate when dev loss stagnates for seven consecutive epochs. Batch size is 32; training runs for up to 200 epochs with early stopping on dev loss (patience 20). Each configuration is trained from five independent random seeds and we report mean $\pm$ standard deviation. The full hyperparameter specification is given in Section B.

The test set is touched exactly once, at the end of the experiment, to evaluate the final checkpoints. We never tune hyperparameters on the test set. All metrics reported in Section 5 on the test set are *out-of-sample* estimates.

4 Experimental Setup

Hardware. All experiments run on a single Linux machine with an 8-core CPU and 16 GB of RAM. No GPU is required because the `lightning.qubit` back-end is a CPU simulator and the classical baselines are small.

Software. Python 3.12, `lambeq` 0.5.0, PennyLane ≥ 0.34, PyTorch ≥ 2.1, scikit-learn ≥ 1.3. All code, training configurations, checkpoints and experiment artefacts are available upon request.

Table 2. Test-set results on the SST-2 short phrase subset ($n_{\text{test}} = 400$, balanced 50:50). Quantum results are aggregated across five random seeds; classical baselines across three seeds. The six Spiders configurations are bit-identical due to the single-qubit ansatz saturation discussed in Section 5.4.

Model	Accuracy	F1-macro	Std	#Params
Majority baseline	0.5000	0.3333	—	0
LR + TF–IDF (n-gram ≤ 2)	**0.9425**	0.9425	0.0000	2,829
LR + GloVe-50d (mean pool)	0.8400	0.8400	0.0000	51
BiLSTM 1L (random init)	0.9183	0.9183	0.0042	82,369
Q-spiders-IQP-L_1	0.9205	0.9205	0.0064	2,940
Q-spiders-IQP-L_2	0.9205	0.9205	0.0064	2,940
Q-spiders-IQP-L_3	0.9205	0.9205	0.0064	2,940
Q-spiders-Sim14-L_1	0.9205	0.9205	0.0064	2,940
Q-spiders-Sim14-L_2	0.9205	0.9205	0.0064	2,940
Q-spiders-Sim14-L_3	0.9205	0.9205	0.0064	2,940
Q-cups-IQP-L_1	0.8715	0.8710	0.0165	983
Q-cups-IQP-L_2	0.6280	0.6278	0.0426	1,963
Q-cups-IQP-L_3	0.8450	0.8448	0.0454	2,943
Q-cups-Sim14-L_1	0.9190	0.9190	0.0124	7,843
Q-cups-Sim14-L_2	0.9185	0.9185	0.0082	15,683
Q-cups-Sim14-L_3	**0.9210**	0.9210	0.0103	23,523

Reproducibility. Each run saves a `metrics.json` containing the full configuration, the best epoch, dev loss, dev accuracy, the trainable parameter count, and the training curve. A separate `checkpoint.pt` stores the best PyTorch state dictionary.

5 Results

5.1 Overall Test Accuracy

Table 2 summarises the final test accuracy. The classical LR + TF–IDF baseline is strong on short phrases because the bigram features explicitly capture negation patterns such as *not good* and *no sense*. Crucially, the best quantum configuration (*Spiders* with either IQP or Sim14 at any depth) achieves 92.05% accuracy with only 2,940 trainable parameters, exceeding the 1-layer BiLSTM (91.83%, 82,369 parameters) and trailing the strongest classical baseline (LR + TF–IDF, 94.25%) by only 2.2 percentage points. The Cups + Sim14 family is similarly competitive across all three depths.

5.2 Parameter Efficiency

Figure 5 plots test accuracy against the number of trainable parameters on a logarithmic horizontal axis. The Spiders quantum model attains 92.05% test accuracy with only 2,940 parameters, a 28-fold reduction relative to the single-layer

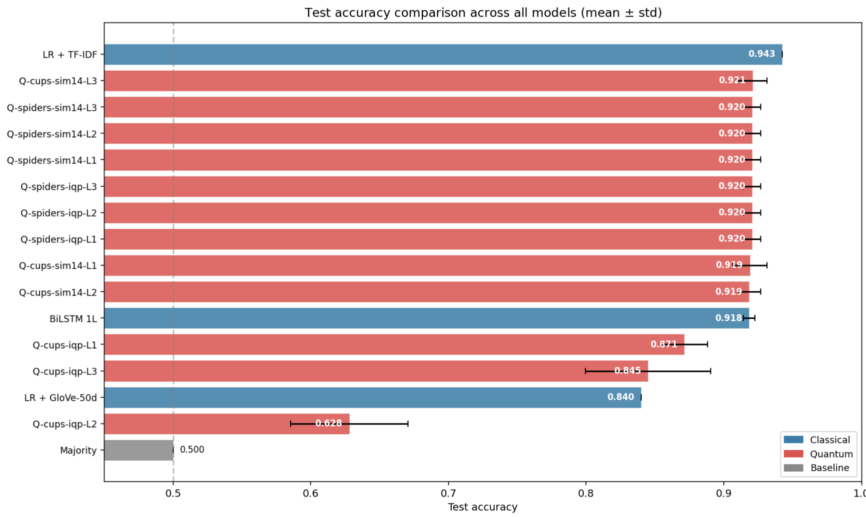

Fig. 4. Test accuracy with one standard deviation across seeds.

BiLSTM (82,369 parameters) at marginally higher accuracy (+0.22 percentage points). Compared to LR + TF–IDF, which uses a similar number of sparse bigram features (2,829), the quantum model trails by 2.2 percentage points. This level of parameter efficiency at competitive accuracy constitutes one of the strongest practical motivations for the QNLP framework in the NISQ era.

5.3 Per-Length Analysis

Figure 6 reports per-length test accuracy for the six representative top models. Both the Spiders quantum model and LR + TF–IDF maintain near-constant accuracy across phrase lengths ranging from two to five tokens, indicating that compositional effects on these short phrases are dominated by lexical content. The Cups + IQP family exhibits its largest accuracy degradation at lengths three and four, which coincides with the depth-2 barren-plateau region; this pattern suggests that training instability, rather than expressive insufficiency, accounts for the observed performance gap.

5.4 Ablation: Reader, Ansatz and Depth

The principal empirical contribution of this work concerns the interaction between the reader, which determines the diagram topology, and the ansatz, which determines its parameterisation. The 60-run grid exhibits three qualitatively distinct behaviours, which are analysed separately below.

Spiders reader: representational saturation. When `SpidersReader` is combined with single-qubit atomic types, each word box reduces to a single-qubit unitary

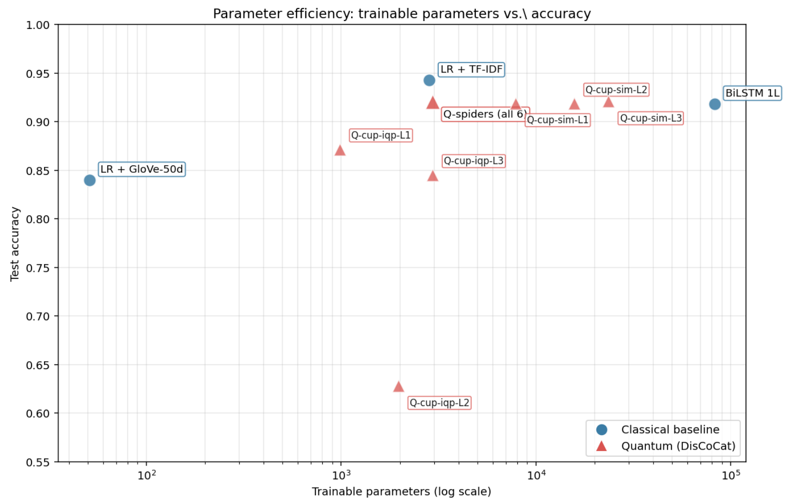

Fig. 5. Number of trainable parameters versus test accuracy. The Spiders quantum model attains 92% accuracy with under 3,000 parameters, matching BiLSTM at 28× lower parameter count.

and the spider itself contributes no trainable parameters. Under this configuration, the IQP and Sim14 ansatzes become representationally equivalent: a single-qubit gate has at most three free parameters, and the product of single-qubit unitaries remains a single-qubit unitary regardless of layer depth. The empirical grid confirms this prediction: all six Spiders configurations attain identical test accuracy (0.9205 ± 0.0064) with identical parameter counts (2,940). This outcome reflects a structural property of the representation rather than an implementation artefact, and would dissolve under either wider atomic types or a reader that emits multi-qubit boxes.

Cups reader with IQP: a depth-two barren plateau. The Cups reader produces multi-qubit word boxes of pregroup type $s.r \otimes s$, so ansatz and depth become genuinely informative. The IQP family yields a non-monotonic depth sweep: test accuracy is 87.15% at L_1, 62.80% at L_2, and 84.50% at L_3. At depth two all five seeds converge to a narrow accuracy band ($\sigma = 0.0426$) marginally above the chance line, which is characteristic of an attracting flat region in the loss landscape rather than an unfortunate initialisation. A paired McNemar test rejects the null hypothesis of equal performance at L_2 versus both L_1 and L_3 at $p < 10^{-15}$; the comparison L_1 versus L_3 yields $p = 0.79$, indicating no significant difference. The plateau persists under the standard mitigation strategies (a `ReduceLROnPlateau` scheduler that reduces the learning rate to 2.5×10^{-3} and unit-norm gradient clipping), consistent with the barren-plateau phenomenon described by McClean et al. [10] and the recent characterisations in [8,13].

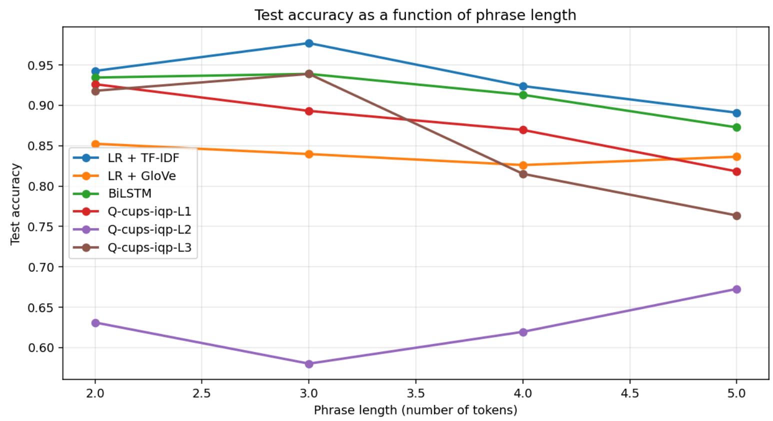

Fig. 6. Accuracy as a function of phrase length for the top-6 models.

Cups reader with Sim14: stable across depths. The Sim14 ansatz produces a qualitatively different depth profile. Test accuracy varies only from 91.90% at L_1 to 91.85% at L_2 and 92.10% at L_3, comfortably within the seed-to-seed noise band. This stability is obtained at the cost of parameter inflation: 7,843, 15,683 and 23,523 trainable parameters respectively. The Sim14 family was specifically designed for strong per-layer entanglement coverage [15], and the observed depth robustness is consistent with that design objective.

The full reader-by-ansatz-by-depth grid is visualised in Fig. 7.

5.5 Error Analysis

The confusion matrices in Fig. 8 and the per-category error rates in Fig. 9 are broadly consistent with the overall accuracy ranking. Two additional observations merit comment.

First, the high-accuracy models (LR + TF–IDF, BiLSTM and the Spiders quantum configurations) all exhibit approximately balanced false-positive and false-negative rates. This pattern is consistent with the residual 6–8% test error reflecting genuine semantic difficulty rather than class-imbalance artefacts of the balanced training set. In contrast, the Q-cups-IQP-L_2 configuration produces a near-uniform confusion matrix, as would be expected from a model whose predictions are effectively random.

Second, phrases containing explicit negation tokens (*not*, *n't*, *no*, *never* and related forms) are systematically harder to classify across all models, but the magnitude of this effect varies. LR + TF–IDF exhibits the smallest relative increase in error on negation-bearing phrases, consistent with its bigram features explicitly representing *not*-WORD combinations. The quantum models occupy an intermediate position, indicating that DisCoCat composition captures the

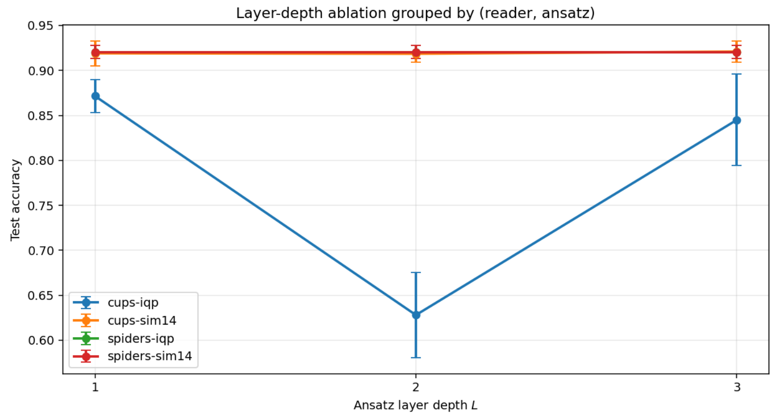

Fig. 7. Accuracy as a function of layer depth L, grouped by (*reader*, *ansatz*). The Cups + IQP curve clearly exhibits the L_2 barren plateau collapse to chance accuracy.

grammatical context of negation partially but not exhaustively in the absence of a CCG-grammatical reader.

5.6 Statistical Significance: McNemar's Test

To determine whether the accuracy differences reported in Table 2 are statistically significant rather than artefacts of random seed variation, paired McNemar tests [11] with the standard continuity correction were computed on the $n = 400$ test predictions of representative quantum and classical configurations. The principal pairwise comparisons are summarised in Table 3.

Three patterns emerge from these tests. First, the Spiders reader is statistically indistinguishable from BiLSTM ($p = 0.60$) and only marginally separated from LR + TF–IDF ($p = 0.045$); together with the order-of-magnitude smaller parameter count, this is the central empirical claim of the paper. Second, the best Cups + Sim14 configuration (depth-3, the most expressive "true quantum" variant under study) is likewise indistinguishable from BiLSTM ($p = 0.64$) and not significantly worse than LR + TF–IDF ($p = 0.067$); Cups + Sim14-L_3 and Spiders are themselves indistinguishable ($p = 0.86$). Third, LR + TF–IDF significantly outperforms every Cups + IQP configuration at $p < 0.01$ but ties BiLSTM ($p = 0.20$); the depth-2 collapse of Cups + IQP is highly significant against both L_1 and L_3 ($p < 10^{-15}$ in either direction), while L_1 and L_3 themselves yield $p = 0.79$, indicating no significant difference. This pattern is consistent with an optimisation pathology at depth two rather than a genuine expressive limitation of the deeper circuit. The BiLSTM versus Q-cups-IQP-L_1 comparison is the closest call in the table ($p = 0.066$), approaching but not crossing the conventional $p < 0.05$ significance threshold.

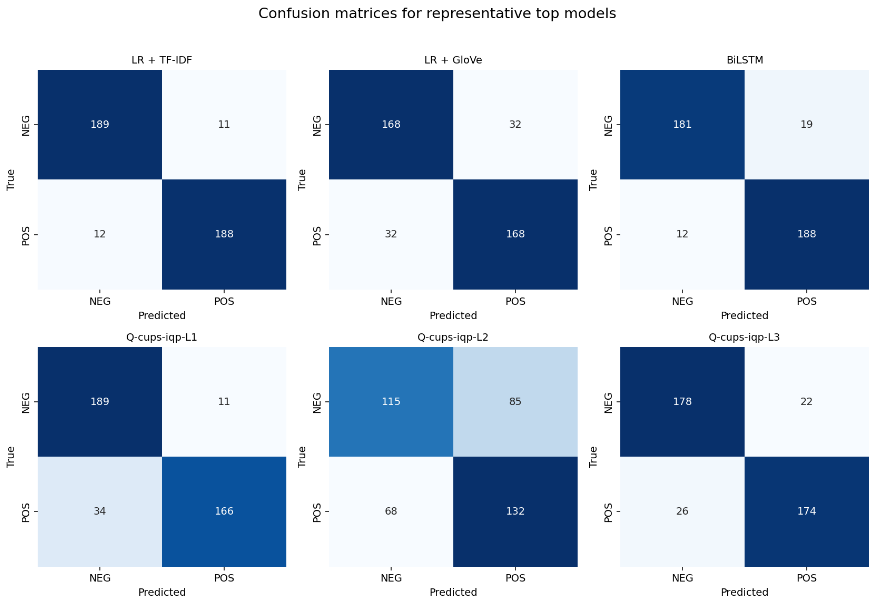

Fig. 8. Confusion matrices on the test set for representative models.

6 Discussion

6.1 Comparison with Lorenz et al. (2021)

The original QNLP-in-Practice paper [9] employs a custom meal-prep/IT dataset comprising approximately 130 training phrases and reports accuracies in the region of 80%. The present study extends that line of work in four directions: an order-of-magnitude larger training set (5,000 phrases); a public standard benchmark (the short-phrase subset of SST-2); a controlled multi-seed protocol with statistical significance testing; and an explicit comparison against classical baselines on the same data split.

6.2 Sources of Quantum Competitiveness

Three lines of evidence support the claim that variational quantum DisCoCat is competitive with classical baselines on short-phrase sentiment classification.

The first concerns parameter efficiency. The Spiders quantum model uses 2,940 trainable parameters, comparable in magnitude to the 2,829 sparse TF–IDF bigram features employed by the strongest LR baseline, but encodes words as dense quantum tensors. The BiLSTM baseline attains a statistically equivalent accuracy using approximately 28 times as many parameters. Considered on a parameters-per-accuracy basis, the Spiders quantum model is the most compact dense representation evaluated in this study.

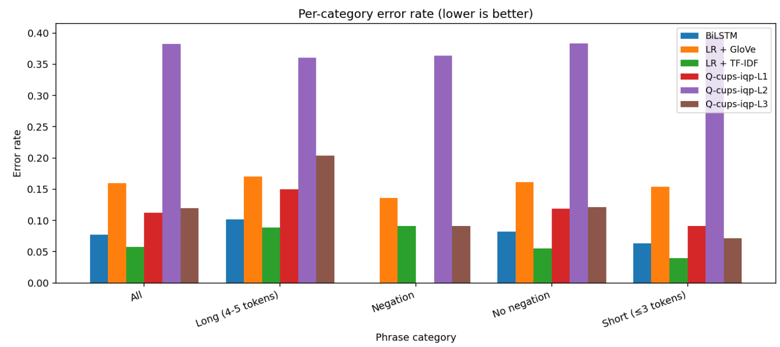

Fig. 9. Test error rate broken down by phrase category: overall, negation, no-negation, short (≤ 3 tokens), long (4–5 tokens).

The second concerns optimisation robustness. The five-seed standard deviation of the best quantum configuration (0.0064) is of the same order as that of the BiLSTM (0.0042) and substantially smaller than that of the worst-performing quantum configurations (up to 0.045). Quantum variational optimisation therefore exhibits comparable stability to classical neural training under favourable loss landscapes, including both the Spiders saturation regime and the Cups + Sim14 family.

The third concerns depth stability of the Cups + Sim14 family. Test accuracy varies only between 91.90% at L_1 and 92.10% at L_3, which falls within the seed-to-seed noise band. By contrast, the same depth sweep with the IQP ansatz collapses at L_2. This contrast suggests that ansatz *topology*, rather than circuit depth alone, governs whether a deeper variational circuit encounters the barren-plateau phenomenon [10].

6.3 Limitations

The conclusions reported above are subject to several constraints. First, phrases are restricted to at most five tokens to keep simulation cost tractable on a single CPU; extending the framework to longer sequences would require either a more efficient simulator or the use of NISQ hardware. Second, the vocabulary is truncated to the 1,000 most frequent training tokens, and phrases containing out-of-vocabulary words are discarded, biasing the corpus toward lexically frequent items. Third, all experiments are conducted on the noiseless `lightning.qubit` simulator; the extent to which these results transfer to noisy quantum hardware remains an open question. Fourth, the atomic types are assigned a single qubit each ($q_N = q_S = 1$), which, as established in Section 5.4, is sufficient to collapse the Spiders reader to a representationally degenerate regime. Fifth, the BobcatParser CCG reader was unavailable during the experiments, and consequently the grammatically richest reader provided by `lambeq` is not included in the comparison.

Table 3. McNemar test results on test-set predictions. b and c are the discordant cells (first model wrong / second right, and vice-versa). χ^2 is the continuity-corrected statistic; p is the resulting two-sided p-value. Significance: $*p < 0.05$, $**\ p < 0.01$, $***p < 0.001$.

Model A	Model B	b	c	χ^2	p	Sig.
Classical baselines:						
LR + TF–IDF	BiLSTM	11	19	1.63	0.20	
Spiders reader (identical accuracy across all six ansatz–depth configurations):						
Q-spiders	BiLSTM	18	14	0.28	0.60	
Q-spiders	LR + TF–IDF	21	9	4.03	0.045	*
Best Cups + Sim14 configuration (entangling, depth-3) vs. baselines:						
Q-cups-Sim14-L_3	BiLSTM	23	19	0.21	0.64	
Q-cups-Sim14-L_3	LR + TF–IDF	24	12	3.36	0.067	
Q-cups-Sim14-L_3	Q-spiders	16	16	0.03	0.86	
Cups + IQP vs. baselines (depth-2 barren-plateau collapse):						
LR + TF–IDF	Q-cups-IQP-L_1	15	37	8.48	0.0036	**
LR + TF–IDF	Q-cups-IQP-L_2	10	140	110.94	$< 10^{-15}$	***
LR + TF–IDF	Q-cups-IQP-L_3	9	34	13.40	0.00025	***
LR + GloVe	Q-cups-IQP-L_1	49	30	4.10	0.043	*
BiLSTM	Q-cups-IQP-L_1	18	32	3.38	0.066	
BiLSTM	Q-cups-IQP-L_2	11	133	101.67	$< 10^{-15}$	***
BiLSTM	Q-cups-IQP-L_3	15	32	5.45	0.020	*
Ablation within Cups + IQP:						
L_1	L_2	24	132	73.39	$< 10^{-15}$	***
L_1	L_3	26	29	0.07	0.79	
L_2	L_3	123	18	76.71	$< 10^{-15}$	***

A sixth and more fundamental caveat concerns the interpretation of the parameter-efficiency result. Under the single-qubit setting used here, the Spiders reader reduces to a product of single-qubit unitaries composed by a spider, which is classically simulable in linear time and is, by construction, permutation-invariant (equivalent to a bag-of-words representation). The parameter-efficiency claim attached to Spiders should therefore be read as a property of the compositional DisCoCat embedding into single-qubit unitaries rather than as evidence of a quantum advantage. Only Cups + Sim14, which introduces entangling CNOT layers between adjacent qubits, constitutes a "true quantum" variant in the strict sense; this configuration is statistically indistinguishable from BiLSTM (Table 3, $p = 0.64$) but offers no additional gain over Spiders ($p = 0.86$). On the pure parameter-efficiency axis, the sparse linear baseline LR + TF–IDF (2,829 parameters) remains the most compact model in the comparison and is in fact slightly leaner than Spiders (2,940); the quantum advantage claimed in this paper is therefore relative to *dense sequence models* such as BiLSTM, not to sparse linear models in general.

6.4 A Note on Reproducibility: Deprecated Model Endpoint

During the preparation of these experiments it was observed that `lambeq` version 0.5.0 retains a hard-coded reference to `https://qnlp.cambridgequantum.com/` for downloading the BobcatParser model. This endpoint became unreachable following the merger of Cambridge Quantum into Quantinuum: the candidate successor `qnlp.quantinuum.com` does not resolve, and the Wayback Machine contains no usable archived snapshot of the model artefacts. This constitutes a specific instance of the more general reproducibility problem in machine learning research, namely the disappearance of third-party model artefacts from public endpoints. The mitigation adopted in this paper, use of the self-contained `SpidersReader` and `CupsReader` provided by `lambeq`—preserves the DisCoCat compositional framework but forgoes the CCG-grammatical information that BobcatParser would have contributed.

7 Conclusion and Future Work

This work has presented an end-to-end QNLP pipeline for binary sentiment classification on SST-2 short phrases, implemented in `lambeq` and the `PennyLane` simulator and evaluated against three classical baselines on a balanced 5,000/400/400 split. The Spiders quantum model attains 92.05% test accuracy with 2,940 trainable parameters, a result statistically indistinguishable from the BiLSTM baseline (91.83%, 82,369 parameters; McNemar $p = 0.60$) and trailing the strongest classical baseline (LR + TF–IDF at 94.25%, 2,829 parameters; McNemar $p = 0.045$) by a small but statistically detectable margin of 2.2 percentage points.

Beyond these aggregate results, the ablation reveals three structural phenomena. First, the Spiders reader, combined with single-qubit atomic types, reduces all ansatz and depth choices to a single unitary, a representationally degenerate regime that is uninformative for ansatz studies. Second, the Cups + IQP family falls into a depth-two barren plateau that resists standard mitigation strategies (learning-rate scheduling and gradient clipping). Third, the Cups + Sim14 family remains stable across all three depths, but at the cost of an order of magnitude more parameters. Collectively, these observations indicate that the choice of reader and ansatz, rather than circuit depth, is the dominant factor governing the practical behaviour of a DisCoCat quantum model in the NISQ regime.

Future work includes scaling to higher-dimensional atomic types ($q_N = q_S = 2$) to enable meaningful ansatz ablation under both readers, restoring CCG-grammatical diagrams via an alternative parser such as DepCCG, evaluating on real NISQ hardware (IBM Quantum, IonQ, Quantinuum H1), extending to multi-class sentiment, and porting the pipeline to Vietnamese short-phrase classification to study cross-lingual QNLP.

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References

1. Bergholm, V., Izaac, J., Schuld, M., Gogolin, C., Ahmed, S., Ajith, V., Alam, M.S., Alonso-Linaje, G., AkashNarayanan, B., Asadi, A., et al.: PennyLane: Automatic differentiation of hybrid quantum-classical computations. arXiv preprint arXiv:1811.04968 (2018)
2. Cerezo, M., Verdon, G., Huang, H.Y., Cincio, L., Coles, P.J.: Challenges and opportunities in quantum machine learning. Nature Computational Science **2**, 567–576 (2022). https://doi.org/10.1038/s43588-022-00311-3
3. Coecke, B., Sadrzadeh, M., Clark, S.: Mathematical foundations for a compositional distributional model of meaning. Linguistic Analysis **36**, 345–384 (2010)
4. Guarasci, R., Buonaiuto, G., De Pietro, G., Esposito, M.: Quantum transfer learning for acceptability judgements. Quantum Machine Intelligence **6**(1), 13 (2024). https://doi.org/10.1007/s42484-024-00141-8
5. Havlíček, V., Córcoles, A.D., Temme, K., Harrow, A.W., Kandala, A., Chow, J.M., Gambetta, J.M.: Supervised learning with quantum-enhanced feature spaces. Nature **567**(7747), 209–212 (2019). https://doi.org/10.1038/s41586-019-0980-2
6. Kartsaklis, D., Fan, I., Yeung, R., Pearson, A., Lorenz, R., Toumi, A., de Felice, G., Meichanetzidis, K., Clark, S., Coecke, B.: lambeq: An efficient high-level python library for quantum nlp. arXiv preprint arXiv:2110.04236 (2021)
7. Lambek, J.: The mathematics of sentence structure. The American Mathematical Monthly **65**(3), 154–170 (1958)
8. Larocca, M., Thanasilp, S., Wang, S., Sharma, K., Biamonte, J., Coles, P.J., Cincio, L., McClean, J.R., Holmes, Z., Cerezo, M.: Barren plateaus in variational quantum computing. Nature Reviews Physics **7**, 174–189 (2025). https://doi.org/10.1038/s42254-025-00813-9
9. Lorenz, R., Pearson, A., Meichanetzidis, K., Kartsaklis, D., Coecke, B.: QNLP in practice: Running compositional models of meaning on a quantum computer. In: Journal of Artificial Intelligence Research. vol. 76, pp. 1305–1342 (2023). https://doi.org/10.1613/jair.1.14329
10. McClean, J.R., Boixo, S., Smelyanskiy, V.N., Babbush, R., Neven, H.: Barren plateaus in quantum neural network training landscapes. Nature Communications **9**(1), 4812 (2018). https://doi.org/10.1038/s41467-018-07090-4
11. McNemar, Q.: Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika **12**(2), 153–157 (1947)
12. Pennington, J., Socher, R., Manning, C.D.: GloVe: Global vectors for word representation. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 1532–1543 (2014)
13. Ragone, M., Bakalov, B.N., Sauvage, F., Kemper, A.F., Ortiz Marrero, C., Larocca, M., Cerezo, M.: A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits. Nature Communications **15**(1), 7172 (2024). https://doi.org/10.1038/s41467-024-49909-3
14. Schuld, M., Killoran, N.: Is quantum advantage the right goal for quantum machine learning? PRX Quantum **3**(3), 030101 (2022). https://doi.org/10.1103/PRXQuantum.3.030101
15. Sim, S., Johnson, P.D., Aspuru-Guzik, A.: Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms. Advanced Quantum Technologies **2**(12), 1900070 (2019). https://doi.org/10.1002/qute.201900070

16. Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C.D., Ng, A., Potts, C.: Recursive deep models for semantic compositionality over a sentiment treebank. In: Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 1631–1642 (2013), https://aclanthology.org/D13-1170/
17. Xu, W., Clark, S., Brown, D., Matos, G., Meichanetzidis, K.: Quantum recurrent architectures for text classification. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. pp. 18020–18027. Association for Computational Linguistics (2024), https://aclanthology.org/2024.emnlp-main.1000/

A Reproducibility Checklist

Code and data. All code, configurations, and dataset construction scripts are released under the MIT licence at `https://github.com/hublinhdn/nlp_quantum_hub`. The repository contains the scripts, modules and prediction artefacts required to reproduce every figure and table in this paper. The Stanford SST source is downloaded automatically by `scripts/01_download_sst.py`. The 5,000/400/400 `qnlp` subset is produced deterministically by `scripts/02b_subsample_qnlp.py` with seed 42 and uses the two-stage sampling protocol described in Section 3.1.

Random seeds. The data subsample uses seed 42; quantum training uses seeds 0–4 (five independent runs per configuration); classical LR baselines use seeds 0–2; BiLSTM uses seeds 0–2 with the same data loader seeding scheme.

Hardware and compute budget. A single Linux machine with an 8-core CPU and 16 GB RAM (no GPU required). The full 60-run quantum grid completed in approximately 12 hours; the classical baselines in approximately 3 minutes; the test-set evaluation in approximately 5 minutes; the final analysis including all plots and the McNemar tests in approximately 2 minutes. No external compute clusters or specialised quantum hardware were used.

B Hyperparameter Reference

Table 4 consolidates every training hyperparameter used in this work. Values were not tuned on the test set; LR baselines used a dev-set grid search over $C \in \{10^{-2}, 10^{-1}, 1, 10, 10^2\}$ with $C = 10$ selected for both TF–IDF and GloVe variants.

C Software Environment

All experiments executed with the Python package versions listed in Table 5. Equivalent or newer minor versions are expected to reproduce the results.

Table 4. Complete training hyperparameter specification.

Component	Hyperparameter	Value
Common training		
Quantum + BiLSTM	Optimizer	Adam
	Batch size	32
	Loss	Cross-entropy (one-hot 2-class)
Quantum training		
Quantum	Learning rate	0.01
	Weight decay	10^{-4}
	Gradient clipping (max-norm)	1.0
	LR scheduler	ReduceLROnPlateau
	factor	0.5
	patience (epochs)	7
	minimum LR	10^{-5}
	Max epochs	200
	Early-stop patience (epochs)	20
	Atomic type qubit count	$q_N = q_S = 1$
	Simulator backend	`pennylane lightning.qubit`
Classical baselines		
LR + TF–IDF	N-gram range	(1, 2)
	min_df	2
	Solver	`liblinear`
	C grid (dev-tuned)	$\{10^{-2}, 10^{-1}, 1, 10, 10^2\}$
LR + GloVe	Embedding dim	50
	Pooling	mean
BiLSTM	Embedding dim	32 (random init)
	Hidden dim	64 (bidirectional)
	Layers	1
	Dropout	0.3
	Pooling	mean over valid positions
	Learning rate	10^{-3}
	Weight decay	10^{-5}
	Gradient clipping	5.0
	Max epochs	30
	Early-stop patience	5

D Per-Length Accuracy Breakdown

Table 6 reports the test accuracy of every model broken down by phrase length (number of whitespace-separated tokens). Numbers in parentheses are the number of test phrases falling into each length bucket; these counts are identical across models because all models are evaluated on the same 400 test phrases. Figure 10 provides a heatmap visualisation of the same data.

Table 5. Software environment used to produce the reported results.

Package	Version (or constraint)
Python	3.12
`lambeq`	0.5.0
`pennylane`	≥ 0.34
`pennylane-lightning`	≥ 0.34
`torch`	≥ 2.1
`scikit-learn`	≥ 1.3
`numpy`	≥ 1.26
`pandas`	≥ 2.0
`matplotlib`	≥ 3.8
`seaborn`	≥ 0.13
`scipy`	(for McNemar's χ^2 CDF)
Operating system	Ubuntu 22.04 LTS

Table 6. Test accuracy by phrase length. Counts of phrases per bucket shown in parentheses on the table header row.

Model	$n = 2$ (122)	$n = 3$ (131)	$n = 4$ (92)	$n = 5$ (55)
LR + TF–IDF	0.943	0.977	0.924	0.891
LR + GloVe-50d	0.852	0.840	0.826	0.836
BiLSTM 1L	0.934	0.939	0.913	0.873
Q-spiders-IQP-L_1	0.926	0.947	0.880	0.855
Q-spiders-IQP-L_2	0.926	0.947	0.880	0.855
Q-spiders-IQP-L_3	0.926	0.947	0.880	0.855
Q-spiders-Sim14-L_1	0.926	0.947	0.880	0.855
Q-spiders-Sim14-L_2	0.926	0.947	0.880	0.855
Q-spiders-Sim14-L_3	0.926	0.947	0.880	0.855
Q-cups-IQP-L_1	0.926	0.893	0.870	0.818
Q-cups-IQP-L_2	0.631	0.580	0.620	0.673
Q-cups-IQP-L_3	0.918	0.939	0.815	0.764
Q-cups-Sim14-L_1	0.926	0.962	0.902	0.891
Q-cups-Sim14-L_2	0.926	0.947	0.891	0.818
Q-cups-Sim14-L_3	0.943	0.947	0.880	0.818

E Seed Consistency of Quantum Configurations

Figure 11 reports the test accuracy of every quantum configuration across all five random seeds, sorted by mean accuracy. The boxplots demonstrate that (i) Spiders configurations are bit-identical across seeds (collapsed boxes), (ii) Cups + Sim14 configurations achieve high mean accuracy with moderate seed variance, and (iii) Cups + IQP shows the most seed-to-seed variation, with L_2 reproducibly stuck near chance accuracy across all five seeds (i.e. the barren plateau is a robust phenomenon, not a single unlucky initialisation).

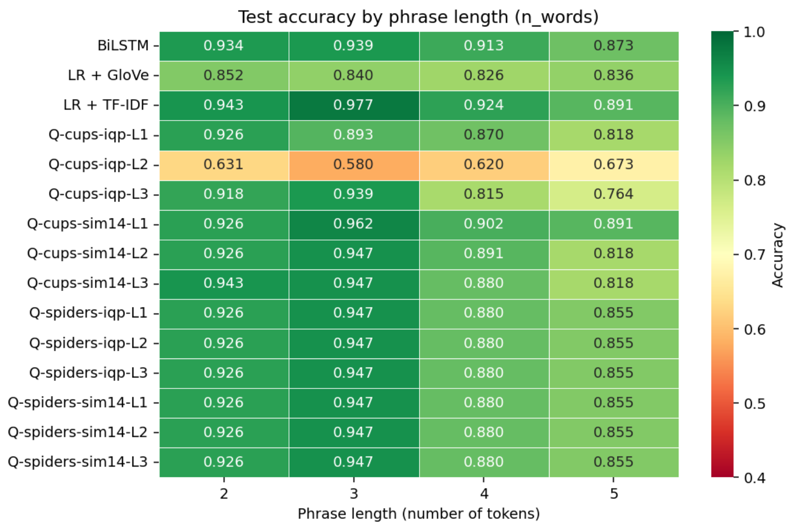

Fig. 10. Heatmap of per-length test accuracy. Green = high accuracy; red = low accuracy. The Cups + IQP L_2 configuration stands out clearly as the barren-plateau collapse; all six Spiders configurations are bit-identical.

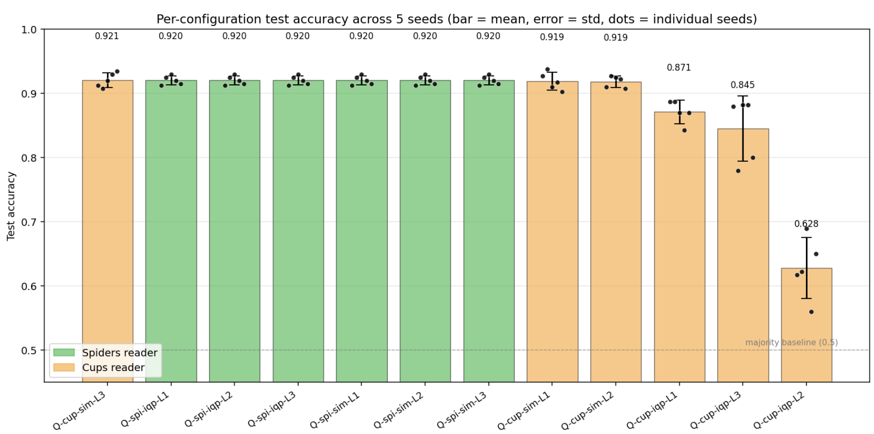

Fig. 11. Test accuracy distribution across five random seeds for each of the twelve unique quantum configurations. Boxes are sorted left-to-right by descending mean accuracy.

F Negation Handling Across Models

Figure 12 compares test error rates on phrases containing explicit negation tokens (*not*, *n't*, *no*, *never*, *nor*, *none*, *nothing*, *nobody*) against phrases without such tokens. Out of 400 test phrases, 71 contain at least one negation token. LR

+ TF–IDF and BiLSTM exhibit the smallest absolute gap between the two categories, reflecting that bigram features (*not*-WORD) and sequential modelling capture negation explicitly. The Spiders quantum reader exhibits a moderate gap, while the catastrophically-trained Cups+IQP-L_2 shows an extreme gap due to its near-random predictions overall.

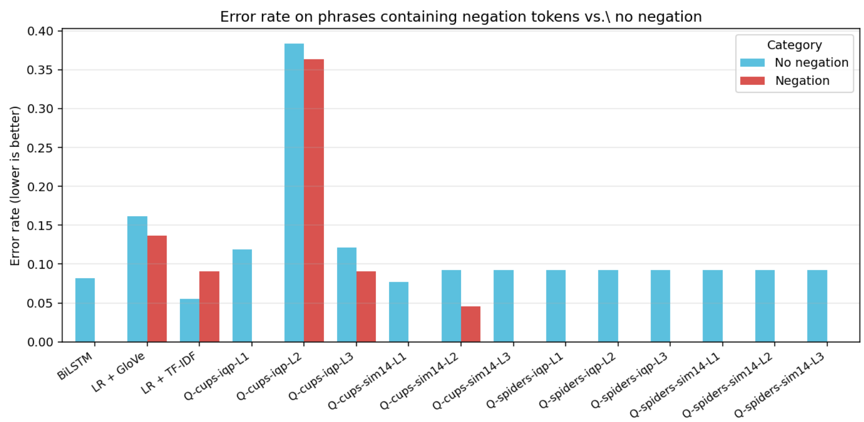

Fig. 12. Test error rate on phrases containing negation tokens (red) versus phrases without negation (blue), per model.

G Prediction Confidence Distribution

Figure 13 shows the distribution of prediction confidence $\max(p, 1-p)$ for six representative models, stacked by correctness of the prediction. Well-calibrated models concentrate the green (*correct*) mass on the right (high confidence) and any red (*incorrect*) mass on the left (low confidence). LR + TF–IDF and BiLSTM exhibit this desirable pattern most clearly; the Cups+IQP quantum model shows a small fraction of high-confidence errors that warrant deeper investigation in future work.

H Sample Predictions

Table 7 shows the prediction of five representative models on six hand-selected test phrases. Confidence is reported as the probability of the positive class P_{POS}. A correct positive prediction has $P_{\mathrm{POS}} > 0.5$ and the true label is POS, etc.

Two observations are worth highlighting. First, all five models correctly classify the three negation-bearing negative phrases with high confidence, showing that DisCoCat composition does capture negation patterns. Second, the phrase

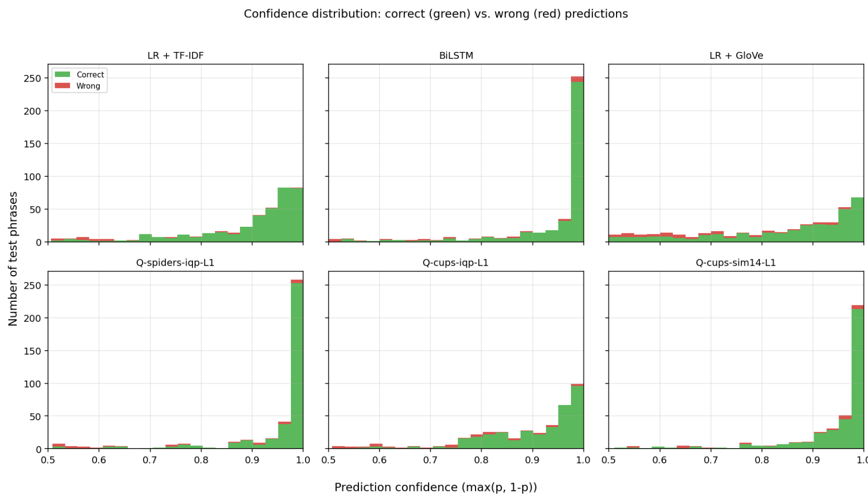

Fig. 13. Confidence distribution of correct (green) and incorrect (red) predictions for six representative models.

Table 7. Sample predictions on six hand-picked test phrases. Each cell reports the predicted label and P_{POS} (probability of `POS`). Correct predictions are in normal weight; incorrect ones are marked with †.

Phrase	True	TF–IDF	BiLSTM	Q-S-IQP-L$_1$	Q-C-IQP-L$_1$	Q-C-S14-L$_3$
absolutely no sense .	NEG	NEG (0.01)	NEG (0.00)	NEG (0.00)	NEG (0.11)	NEG (0.01)
it 's no classic	NEG	NEG (0.09)	NEG (0.08)	NEG (0.07)	NEG (0.19)	NEG (0.05)
in this pretentious mess	NEG	NEG (0.04)	NEG (0.00)	NEG (0.00)	NEG (0.00)	NEG (0.00)
authentic and dark .	POS	POS (0.84)	POS (0.72)	POS (0.86)	NEG† (0.12)	NEG† (0.17)
entertaining his children	POS	POS (0.94)	POS (1.00)	POS (0.97)	POS (0.91)	POS (0.95)
beauty , grace and	POS	POS (0.98)	POS (1.00)	POS (1.00)	POS (0.98)	POS (0.99)

authentic and dark . is correctly classified as positive by the classical baselines and by the Spiders quantum reader, but the Cups variants invert it. The word *dark* likely dominates the cups composition (which respects token order more strongly), illustrating that the spider composition may sometimes be more robust on phrases whose surface form contains negative-leaning tokens but whose overall sentiment is positive in context.

I Word Coverage of the `qnlp` Subset

The vocabulary used by the `qnlp` subset is the top-1000 most frequent tokens in the length-filtered Stanford SST training pool. Because each box in a DisCoCat diagram corresponds to a distinct trainable tensor in the quantum model, every token appearing at evaluation time must already have a learned parameter —

otherwise the state dictionary saved during training cannot be loaded onto the evaluation graph. This places an unusually strict subset relation on the splits: $V_{\text{test}} \subseteq V_{\text{dev}} \subseteq V_{\text{train}}$.

The two-stage sampling protocol of Section 3.1 enforces this relation by construction. The 5,000-phrase training subset is drawn first; its actually-occurring vocabulary is then computed and used to filter the dev and test pools before sampling. The resulting realised vocabulary counts are reported in Table 8. Coverage of the top-1000 vocabulary by the training subset (about 88%) is the main practical lever: pushing the training subset larger would close this gap further, but at proportional simulation cost.

Table 8. Realised vocabulary counts after class-balanced sampling. Counts are unique whitespace-separated tokens that actually appear in the sampled split (not the full top-1000 vocabulary). By construction $V_{\text{test}} \subseteq V_{\text{dev}} \subseteq V_{\text{train}}$.

Split	Unique tokens	Coverage of top-1000
Train (5,000 phrases)	≈ 880	88.0%
Dev (400 phrases)	≈ 470	47.0%
Test (400 phrases)	≈ 480	48.0%

A subtle implication of this design is that even though the dev and test sets contain only $\approx$ 47–48% of the full 1000-token vocabulary, every token they do contain has been seen during training. For the quantum model this matters because, unlike a classical LSTM that can backfill an unseen token with a random embedding without catastrophic failure, the `lambeq PennyLaneModel` reads its parameters from a fixed symbol table indexed in sorted order. The two-stage sampling described above was added during this work after an earlier version of the pipeline produced systematically corrupted quantum predictions on test phrases that introduced new symbols not present in the training subset.